

Cluster Validation

Prerequisite

Students should know about:

- Basics clustering algorithms

Learning Objective

After completing this chapter, student will be able to:

- Explain the statistical cluster validation methods
- Differentiate between Internal and External cluster validation criteria
- List and explain at least two Internal and External validation criteria

Clustering Process

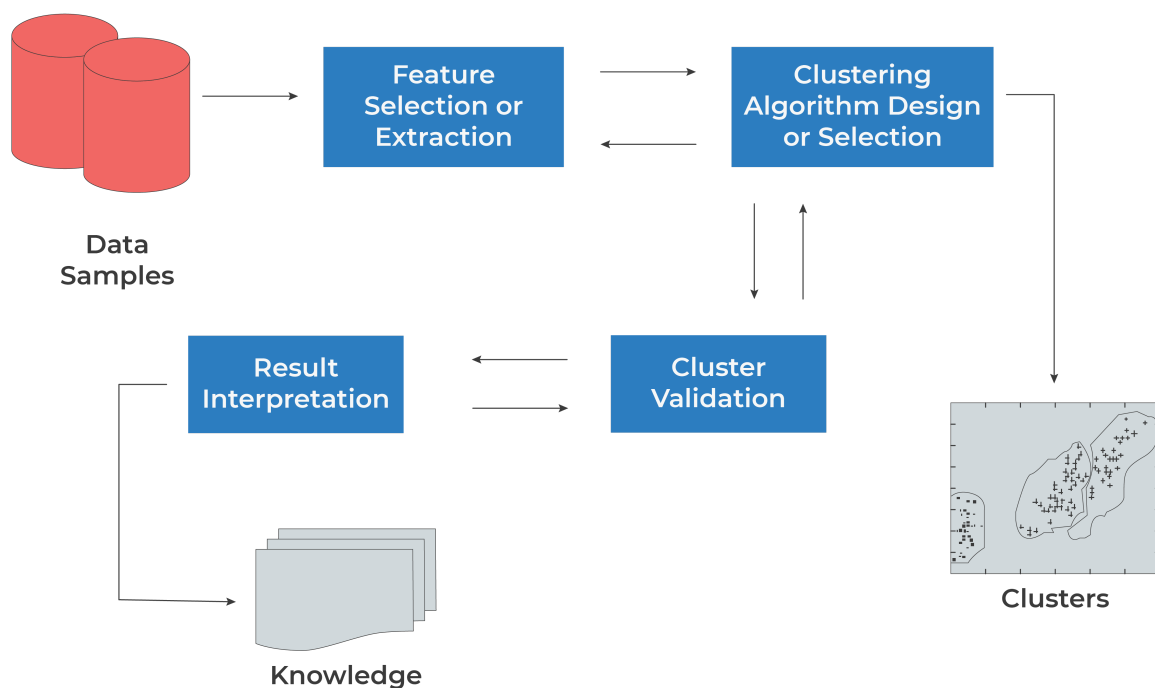


Figure 1: Clustering pipelines (src: Clustering procedure from Xu, R., & Wunsch, D. (2008). *Clustering* (Vol. 10). John Wiley & Sons.)

Figure 1 shows the steps involved in data clustering. Until now, you have learned up to the *Clustering Algorithm Design or Selection* step. As you can see in the figure, we have obtained the clusters, but before we can interpret the result of the clustering, we have to employ the cluster validation. The validation refers to the step of testing the result of clustering. Our goal is to find relevant and important clusters in the data rather than frivolous clusters, so cluster validation is an important step.

Introduction

In the previous chapter you have learnt to perform the clustering for a given dataset using various clustering algorithms. It can be overwhelming to have multiple algorithms. But, having several algorithms allows us to cluster the dataset in many different ways as well as compare their results to select the best method for a dataset. Clustering is an unsupervised learning problem, so it is okay to wonder how can we perform the evaluation when we actually do not have the true result to compare. There are many clever techniques that are used to validate the clustering that you are going to learn in this chapter. Having a set of effective cluster validation skillset will certainly help you identify or choose clusters that are actually present in the data.

Clustering algorithm in itself will always give us the clusters even when the data might not contain the groups or structure. For example, the K -Means algorithm with $K = 2$ will always give us two clusters even though the data may not actually have any grouping in it. With clustering algorithm we are trying to find the non-random structure in the data. If the pattern were actually random, then having that information would also be beneficial for making decisions or extracting knowledge.

In some cases, a domain expert may be able to look at the result produced by a clustering algorithm and evaluate whether the result is cogent or not. But, that will be near impossible when we have a complex and large datasets. So, we need a method that can look at the clustering result and measure the goodness objectively and quantitatively. This will ultimately help us answer the questions like whether the result is actually the non-random structure or just an outcome of the algorithm?

Statistical Method for Clustering Validation

Generally the cluster validation employs a statistical method where we first establish a null hypothesis. The null hypothesis states that the data does not contain any structure. The opposite is the alternative hypothesis that says the data does have an inherent structure. The structure for example can be that the data comes from multiple gaussian distribution with respective parameters or they form grouping of various shapes.

The null hypothesis H_0 can be stated in three way which are given below:

- Random position Hypothesis

The null hypothesis H_0 in this case states that all the locations of the N data points in some specific region of d -dimensional space are equally likely.

- Random graph hypothesis

In random graph hypothesis, H_0 states that all the $N \times N$ order proximity matrices are equally likely.

- Random label hypothesis

The random label hypothesis states the null hypothesis H_0 that all the permutations of the labels on N data points are equally likely.

The random position hypothesis works for ration data, the random graph hypothesis for ordinal proximity between pair of data and the random label hypothesis works for all types of data.

Now, the statistical process of cluster validation can be enumerated as:

1. Establish the null hypothesis stating the absence of structure in the dataset as described above.
2. Select an index that can tell us about the structure of data.

3. Create a distribution of statistic for null hypothesis. This can be a threshold value of the index that separates the acceptance and rejection of hypothesis. The threshold value tells us how small is the "small" and how large is the "large" for our index.
4. Test the hypothesis which indicates whether we accept the null hypothesis or we accept the alternative hypothesis.

These steps seem like a foolproof method for cluster validation, but they are just a statistical process. Even though there are many methods for cluster validation, there does not exist a single conclusive solution. So, the best method for cluster validation remains unknown to us. One method we have already shed light on is to compare the cluster produced by an algorithm with the cluster created by an expert. If the result is similar, we can assume that the clustering algorithm is effective and vice-versa.

In the following section, we will categorize the validation methods and study them individually.

Types of Cluster Validation Criteria

According to the information being used in the validation process, a cluster validation can be categorized into three categories. We will study about these types throughout this chapter. For now, let's understand the basics of those algorithms.

If we have a dataset X and the resulting set of clusters C .

1. Internal Criteria

Internal criteria evaluates the clustering structure from the dataset X without any external information. So, we only utilize the information present in X and C . For example, one of the criteria can be to find out how well separated the clusters are from each other. For this purpose, we only need the information about final cluster.

2. External Criteria

In external criteria, the result of clustering C is compared with some prior information like the predefined category labels P . Even if we have the labels for the dataset, it may not represent the spatial grouping of data, so it might not be a good estimate of clustering.

3. Relative Criteria

In relative criteria, we compare the clustering C with clustering that comes from different clustering algorithms or from the same algorithm with different hyper-parameters. For example, we can try out K -Means with different centroid initialization and compare their results to select the best one.

In the following section we will study some of the validation indices for each cluster validation criteria. The term *criteria* indicates the strategy one should employ in validation, whereas the validation indices represent the quantitative statistics about the clusters with which the clustering can be tested.

As you have already studied, we have to test the hypothesis and make decision. The indices are generally used to test the hypothesis so knowing about them is very important. There are many indices for cluster validation and their effectiveness generally varies with the type of clustering algorithm and the shape of clusters. Here, you will learn about some of the important validation indices for each of the validation criteria.

Internal criteria

The internal criteria uses the clustering result and the dataset to evaluate the quality of the clusters. The objective function of internal criteria can be thought of as placing similar objects in same cluster and dissimilar object in different which is similar to the objective of clustering algorithm. In internal criteria, we have two concepts that can be listed as,

1. Intra-cluster similarity

It signifies the similarity between data points inside a cluster. Having higher intra-cluster similarity insinuates the compactness, connectedness and the homogeneity of the cluster.

2. Inter-cluster similarity

It implies the similarity between two clusters. If two clusters are similar, then they are close to each other and vice-versa.

In this section, we are going to see some of the indices in internal criteria of cluster validation.

Sum of Square Within Clusters (SSW)

If $C = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is a cluster, then the sum of square within cluster can be calculated as,

$$SSW = \sum_{i=1}^n (\mathbf{x}_i - c)^2$$

Where c is the centroid of the cluster calculated as,

$$c = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$$

The SSW can be applied for only the numerical data types because we had to calculate the centroid. It assesses the **compactness** of a given cluster and is suitable if our goal is to obtain spherical clusters.

Sum of Square Between Clusters (SSB)

It measures how well the two given clusters are separated from each other. The separation is measured by calculating the distance of centroid of each clusters to the mean vector of all the data points.

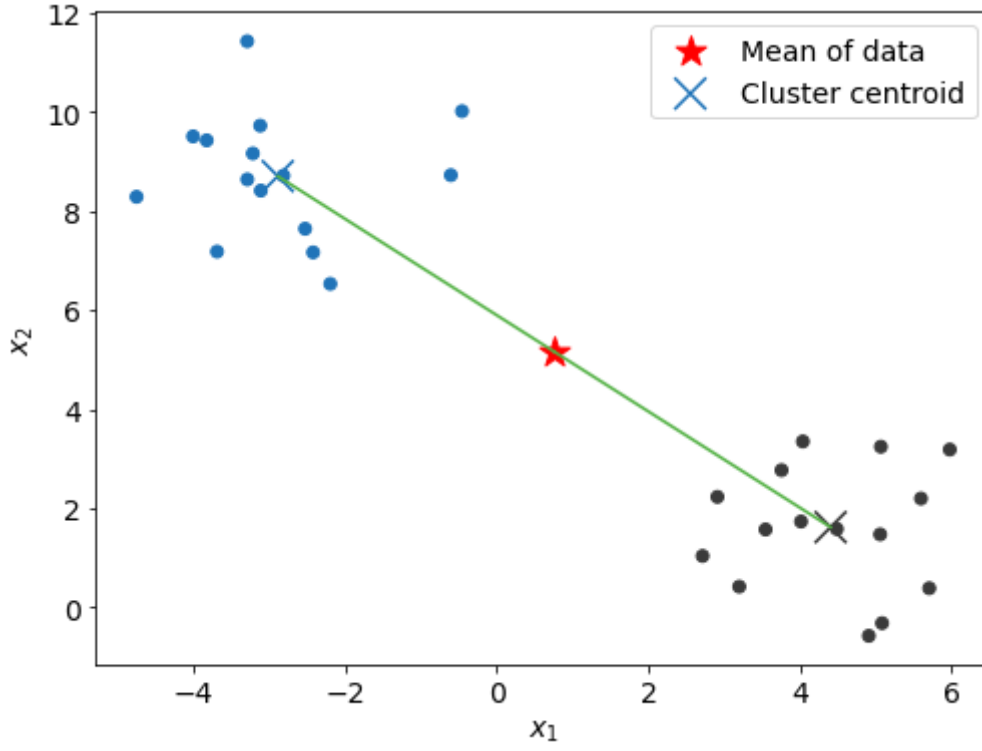


Figure 2: Sum of Square between Clusters

Figure above shows the mean of all the data points, centroid of each clusters, the distance between them. SSB is the sum of distance of centroid to the mean or centroid of all the data points. Higher value of SSB indicates the cluster are well separated.

Silhouette Coefficient (SC)

Silhouette coefficient measures how well the objects of a clusters are placed in it as well as how well they are separated from the objects of different clusters. As you can see, it incorporates both the compactness and separability property we discussed in SSW and SSB.

Let's say $a(x_i)$ calculates the average distance of x_i to other points in same cluster C_k . This value indicates the goodness of assignment for x_i in C_k and also the compactness of that cluster. It can be written as,

$$a(\mathbf{x}_i) = \frac{1}{N_k - 1} \sum_{j=1, j \neq i}^{N_k} d(\mathbf{x}_i, \mathbf{x}_j)$$

Where, N_k is the number of objects in cluster C_k .

Similarly, let's say a function $b(x_i)$ gives us the lowest average dissimilarity of the data point x_i to other clusters. It is given as,

$$b(\mathbf{x}_i) = \min_{k \neq l} \frac{1}{N_l} \sum_{j \in C_l} d(\mathbf{x}_i, \mathbf{x}_j)$$

Now, the silhouette coefficient (SC) is given as,

$$SC(\mathbf{x}_i) = \frac{b(\mathbf{x}_i) - a(\mathbf{x}_i)}{\max\{a(\mathbf{x}_i), b(\mathbf{x}_i)\}}$$

Because we only need the dissimilarity measure $d(\cdot)$ to calculate the SC, it can be applied to any types or data or clustering.

The SC lies between -1 and 1 and more it is closer to 1 the better is the clustering. Below, we compare the SC for two results obtained from K -Means algorithm with different K parameter.

In []:

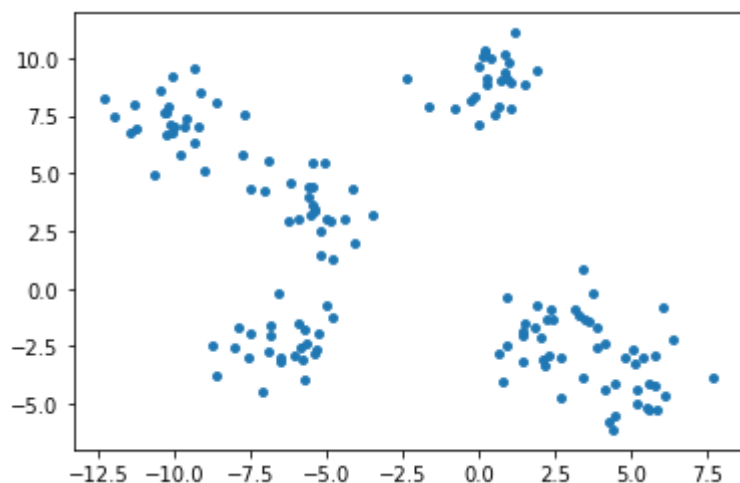
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs

# synthesize the dataset
stds = [0.5, 1.5]

data_labels = []

data, _ = make_blobs(
    150, n_features=2, centers=6, cluster_std=1.1, random_state=23)

plt.scatter(data[:, 0], data[:, 1], s=15)
plt.show()
```



In []:

```
# apply KMeans algorithm
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

random_state = 32
Ks = np.arange(2, 8)

silhouette_scores = []

for K in Ks:
    model = KMeans(n_clusters=K, random_state=random_state)
    # fit model
    model.fit(data)

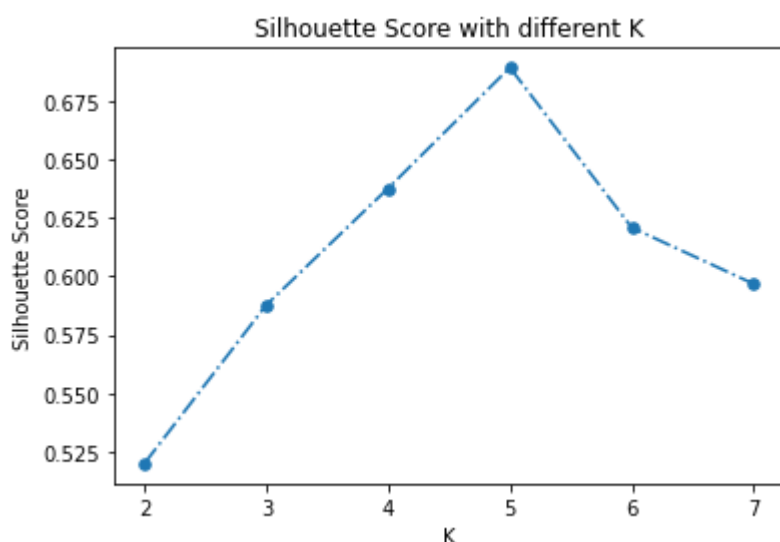
    # calculate silhouette score
    sil_score = silhouette_score(data, model.labels_)
    silhouette_scores.append(sil_score)
    del model

plt.figure(figsize=(6, 4))

plt.plot(Ks, silhouette_scores, '-.')
plt.scatter(Ks, silhouette_scores, s=30, marker='o')
plt.title('Silhouette Score with different K')

plt.xlabel('K')
plt.ylabel('Silhouette Score')

plt.show()
```



In the above plot, you can see that the Silhouette score increases upto a certain value. From this increment we can argue that the compactness as well as the separability of the clusters are getting better. After five clusters, the score starts to fall because the distance between the cluster has decreased (the compactness in *K*-Means always increases when increasing *K*).

Correlation Coefficient

In this measure, we will have two matrices. The first one is a proximity matrix and the second one is an incidence matrix. In the incidence matrix, the value at $I_{i,j}$ is one if the associated pair x_i and x_j belong to the same cluster, otherwise the value is zero.

Now, we will compute the correlation between the proximity matrix and the incidence matrix. Since both the matrices are symmetric, we only have to perform $n(n - 1)/2$ number of calculation. It is considered to be not a good measure for density of contiguity based clustering techniques.

External Criteria

With external criteria, we measure the goodness of clustering result by comparing the clustering to some ground truth (if available) or other clusters independent of the algorithm.

Let's assume we have a predefined category in data X given by P . Then in this criteria, we compare the resulting clustering C with P . We are going to talk about some of the clustering criteria given below:

Rand Index and Jaccard Coefficient

Rand indices and Jaccard coefficient comes under the *pair-counting measures* because we are using the aggrement of counted values for pair of data points in the two clusters. So, let's first define how we count the pairs in the dataset.

If we have two data points x_i and x_j from X , then we can have four different combination about the assignment of those pair of data.

- They both lies in the same cluster in both P and C . Let's denote the number of those aggrements as a .
- They both lies in different cluster in both P and C . This is also the aggrement between the clusters and let's denote this with b .
- The pair of data points lies in same cluster in P but in different cluster in C . Let's denote this disagreement in the cluster assignment with c .
- Also, the pair can be in different clusters in P but in same cluster in C . This is also the disagreement between the cluster assignement and let's denote it using d .

Now, the Rand index can be given as,

$$R = \frac{a + b}{a + b + c + d} = \frac{a + b}{\binom{n}{2}}$$

The rand index gives the normalized value of aggrement between the two data clusters. The numerator indicates the aggrement of the two clusters, whereas the denomenator is the total number of pairs of data available.

Similarly, the **Jaccard coefficient** can be calculated as,

$$J = \frac{a}{a + b + c}$$

Mutual Information (MI)

Mutual Information (MI) falls into the **information theoretic category**. Most of the information theoretic measures uses the concept of entropy when comparing the clusters. MI gives us the information about the reduction of entropy of class labels that we get if we know the cluster labels. Mathematically, it can be written

as,

$$MI = H(P) - H(C|P)$$

Here, $H(P)$ denotes the entropy of class labels. Because we have the class label information even before we apply clustering, this can be calculated even before the clustering. On the other hand, $H(C)$ is the entropy of cluster labels. It is calculated for each of the clustering results. The term $H(C|P)$ is the conditional entropy of class labels P for clustering C .

The MI can be written in terms of probability as,

$$MI(P, C) = \sum_{i=1}^k \sum_{j=1}^l p(P_i, C_j) \log \frac{p(P_i, C_j)}{p(P_i)p(C_j)}$$

Relative Criteria

As you have studied above the internal and external criteria employ the statistical hypothesis testing methodology. The indices were used to either accept or reject the null hypothesis. Relative criteria is a little bit different because it is more of a methodology than statistical approach. The external and internal criteria were also computationally expensive which may not be always the case for relative criteria. The relative criteria compares the result of clustering from different algorithms or from the same algorithm but with different parameters.

One of the issue in clustering is to know the number of clusters in the data which is generally denoted by K . We will use this parameter as an example because it is common in both partition based clustering where we have to prespecify K and in hierarchical clustering where we should know the level where to cut the dendrogram. Underestimating and overestimating the value of K can have undesirable effects. For example, if we underestimate the number of clusters, then the resultant clusters will contain more than one real clusters of data and the interpretation of result would be faulty. Similarly, overestimating the number of clusters will sometimes break a single cluster and the interpretation of result would not be effective. Remember that the algorithm will always produce the specified number of cluster even though there might be lesser or more clusters.

We will discuss two relative criteria below.

1. Data visualization

Data visualization can give us the information about the number of clusters or other parameters if the visualization of data is effective in 2D or 3D space. In general, the data are more complex and contains attribute in dimension greater than three. If the visualization is possible using various technique, we can get some intuition of data and parameters to experiment with.

2. Validation indices

Another method to find the best parameter for example the number of clusters K is to run the algorithm multiple times from say K_{min} to K_{max} , then evaluate some indices to compare their results. Internal indices you studied above can be used to evaluate the result or some objective function. Generally, computing the compactness and separability of clusters is a good way to evaluate the individual results.

The Elbow Method you studied in the K -Means algorithm falls into this type of validation where we performed the operation described above with increasing value of K .

Key Takeaways

- Before using or inferring information from the result of clustering, we have to check the validity of the clustering.
- Clustering validation can be done using statistical hypothesis testing method.
- We have three types of cluster validation criteria according to the information used to validate the clusters.
- Internal criteria uses only the internal informations like data and clustering result, external criteria uses external label and the relative criteria compares the result of two algorithms or single algorithm with different parameter settings.

References

- Books

Aggarwal, C. C., & Reddy, C. K. (2014). Data clustering. Algorithms and applications. Chapman&Hall/CRC Data mining and Knowledge Discovery series, Londra.

- Look at Chapter 23, pages 572-573 for cluster validation basics
- Look at Chapter 23, pages 573-575 for external cluster validation basics
- Look at Chapter 23, pages 589-592 for internal cluster validation basics

Jain, A. K., & Dubes, R. C. (1988). Algorithms for clustering data. Prentice-Hall, Inc..

- Look at Chapter 4, pages 143-147 for cluster validation introduction

Xu, R., & Wunsch, D. (2008). Clustering (Vol. 10). John Wiley & Sons.

- Look at chapter 10, pages 263-265 for introduction to cluster validation
- Look at chapter 10, pages 265-267 for external cluster validation
- Look at chapter 10, pages 267-268 for internal cluster validation
- Look at chapter 10, pages 268 for relative cluster validation